Sonali Mishra | Curriculum Vitae

I-604, Smondo 3 Apartment, Neotown, Electronic City Phase 1, Bangalore, Karnataka, India

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Profile Summary

Motivated and detail-oriented professional with strong expertise in **Artificial Intelligence, IoT, and Cybersecurity**. Experienced in developing and evaluating **AI-based models for cyber threat detection** and network protection. Published multiple papers in reputed journals and conferences, including a **Best Paper Award**. Passionate about applying research-driven problem-solving and analytical skills to deliver secure and innovative technology solutions in a **dynamic IT environment** such as **Bangalore's leading tech organizations**.

Technical and Personal Skills

- **Programming Languages**: Python, MATLAB, HDL
- **AI and Machine Learning Tools:** TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Google Colab, Jupyter Notebook, Databricks
- **Office Automation**: MS PowerPoint, MS Word, MS Excel
- **Internet of Things (IoT)**: IoT Security, IoT Layers, IoT Architecture, IoT Protocols, IoT Devices, Sensors, Actuators, IoT Cyber Security, Raspberry Pi, Arduino
- **Cyber Security**: Penetration Testing, Vulnerability Assessment, Threat Detection, Threat Prevention, Python
- **Cyber Security Tools**: Kali Linux, Wireshark, Image Tool Kit
- **Information Security**: Cryptography, Watermarking, Steganography, AES, DES, RSA, Data Hiding Techniques, MATLAB
- **Data Visualizations**: Tableau
- **Digital Design**: K-map, Combinational Logic Circuits, Sequential Logic Circuits, Boolean
- **Writing:** Report Writing, Technical Paper writing, Book Chapter Writing, Conference paper Writing, Journal Writing, Blog Writing
- **Operating System** Windows, Linux

Publications

1. Sonali Mishra and Ananya Dastidar, "Hybrid image encryption and decryption using cryptography and watermarking for high security application", **IEEE**, **Conference** on Current Trends Towards Converging Technologies (ICCTCT), 2018.

2. Sonali Mishra, Ananya Dastidar, Ipseeta Nanda, "Password protected image encryption and decryption using hybrid technique for data security", **STM Journal** on Current Trends in Signal Processing, Vol.9, ISSN: 2277-6176, 2321-4252, 2019.

3. Sonali Mishra, Ananya Dastidar, Umesh Chandra Samal, Sushant Kumar Mohapatra, "Review of different image data hiding techniques", **International Journal** of Advanced Science and Technology, Vol.121, ISSN: 2005-4238, 2018.

4. Ananya Dastidar and Sonali Mishra, "Encryption and decryption algorithms for IoT device communication", in Electronic devices and circuit design challenges and applications in the Internet of Things book, ed. Suman Lata Tripathi and Smrity Dwivedi (CRC Press: Tylor and Francis Group, 2022), 318. (**Book Chapter**)

5. Sonali Mishra, Anusha K S, Bhuvana G R, Raghunatha Reddy B N, and Mubarak Sharif "Review of Smart ATM Machine", International Journal of All Research Education & Scientific Methods, Vol. 13, Issue 1, ISSN: 2455-6211, 2025.

6. Vivek Gowda G, Kavana Sortur, Sandeep V, Manoj M, and Sonali Mishra, "Asset Monitoring System for Educational Labs: An IoT-based Embedded Hardware–Software Integration", Advanced International Journal for Research (AIJFR), Vol.6, Issue 5, ISSN:3048-7641, 2025.

7. Mrs. Sonali Mishra, Dr. Meena Deshpande, Anjan Kumar S.R, Pavana Kumar G.V, R Yogesh Yadav, and Pritam Rathod, "X-Ray Based Bone Fracture Detection With 2D-3D Reconstruction And Hologhraphic Visualization", International Journal of Creative Research Thoughts(IJCRT), Volume 13, Issue 12, ISSN: 2320-2882, 2025

My Google Sites

- ✓ https://sites.google.com/view/pythonprogramming2ndsem/home
- ✓ https://sites.google.com/view/ddco-bcs302-ise/home
- ✓ https://sites.google.com/view/pythonprograming21ec643/home
- ✓ https://sites.google.com/view/resume-interview/home

Projects

1. AI-Based Resume Analyzer & Skill-Matching Web Application (TalentIQ | Python, NLP, Streamlit)

Technologies: Python, NLP (NLTK), Streamlit, PDF Parsing, Machine Learning Basics, GitHub

- Designed and developed an **AI-powered resume analysis system** using **Python and NLP** techniques.
- Implemented **resume parsing** from PDF files and extracted skills, experience, and keywords using **NLTK**.
- Built **skill-wise match scoring** against job roles with:
  - ✓ Skill match percentage
  - ✓ Missing skills identification
  - ✓ Resume improvement recommendations
- Integrated **skill-based job suggestions** for students and job seekers.
- Developed an **interactive web interface** using **Streamlit** for real-time resume evaluation.
- Deployed the application on **Streamlit Cloud** with GitHub integration.

2. Hybrid AI-Based Cyber Attack Detection & Prevention System (N-BaIoT Dataset)

Technologies: Python, Scikit-learn, Deep Learning, Ensemble Learning, Anomaly Detection

- Developed a hybrid ML–DL detection framework for identifying IoT botnet attacks
- Integrated Random Forest, XGBoost, and Deep Neural Networks
- Achieved high detection accuracy with minimized overfitting
- Implemented stability analysis & statistical validation (t-test / Wilcoxon)
- Generated publication-quality evaluation metrics & visual analytics

3. Intelligent Cyber Threat Detection System (Gowtham Dataset)

Technologies: Python, Machine Learning, Feature Engineering, Classification Models

- Designed AI-based intrusion detection system for smart IoT environments
- Applied supervised ML algorithms for multi-class attack classification
- Optimized hyperparameters for low false positive rates
- Performed comparative performance & robustness analysis

- Developed visual dashboards & contribution analysis

4. Comparative Hybrid AI Security Framework for IoT Networks

Technologies: Python, Ensemble AI, Statistical Modeling, Data Analytics

- Built cross-dataset hybrid AI evaluation framework
- Performed detection stability & statistical significance testing
- Implemented radar charts, ROC analysis, and contribution graphs
- Validated generalization capability across datasets
- Designed publication-ready system architecture & methodology

Certifications and Awards

- ✓ Awarded a certificate of appreciation for presenting a research paper on "Hybrid image encryption and decryption using cryptography and watermarking for high-security application" at the International Conference on Current Trends Towards Converging Technologies (ICCTCT) 2018 under IEEE, Coimbatore, Tamil Nadu.
- ✓ Awarded as best paper for "Password protected hybrid image encryption and decryption process for secure data communication" on VLSI Design, Signal Processing, Image Processing Communication and Embedded Systems (VSPICE) 2020 at Current Trends in Signal Processing under STM Journals, Bhubaneswar, Odisha.
- ✓ Awarded a certificate of appreciation for presenting "Password protected hybrid image encryption and decryption process for secure data communication" at VLSI Design, Signal Processing, Image Processing Communication and Embedded Systems (VSPICE) 2020 under STM Journals, Bhubaneswar, Odisha.
- ✓ Certificate of completion on Internet of Things (IoT) from Central Tool Room and Training Center (CTTC), Bhubaneswar, Odisha.
- ✓ Participated in an online faculty development program on "AI/ML for computer vision and medical image analysis applications" under the Ministry of Electronics and Information Technology (MeitY) 2022, Bengaluru, Karnataka.
- ✓ Certification of completion on IoT from Intellipaat (Online).
- ✓ Certification of completion on Cybersecurity from Simplilearn (Online).
- ✓ Participate in 3-day FDP Workshop on "Implementation of Academic Autonomy", at AMC Engineering College, 2024, Bengaluru, Karnataka, India.
- ✓ Achieved badge Certificate of participation in 'AI For All' program and completed 'AI Appreciate' stage, intel, Digital India 2025.
- ✓ Achieved badge Certificate of participation in 'AI For All' program and completed 'AI Aware' stage, intel, Digital India 2025.

- ✓ Achieved certificate of 'Databricks Accredited Generative AI Fundamentals' from Databricks Academy, 2025.
- ✓ Achieved Google badges on 'Generative AI and Responsible AI', Google, 2025.
- ✓ Achieved Participation certification for attending a 5-day FDP on 'Fundamentals and Application of Edge Computing using Jetson Orin Nano as Edge Device', Sairam College of Engineering, 2025.

Work Experience

Nov/2022 – Feb/2023: **Research Associate at EC-Council, Online mode (Hyderabad, India)**

- Writing different technical papers, academic research, Question Formation, SoP Formation

Dec/2023 – Present: **Assistant Professor, AMC Engineering College, Full-Time (Bangalore, India)**

- Teaching experience on topics of; Python Programming, Digital Design and Computer Organization, Introduction to Electronics and Communication, HDL Programming, MATLAB
- Guided students on the Mini Project "Smart Library using ESP" and Major Project "Smart ATM machine using Retina"
- Mentored students and developed **AI-based applications, including a Resume Analyzer using NLP and Streamlit**.

Educational Qualification

Degree/Board	Institute/School	CGPA/%	Year
PhD (Electronics and Communication Engineering) (**Part Time**)	Reva University, Bangalore, Karnataka	Pursuing	Pursuing (2021)
M.Tech (Electronics and Communication Engineering)	College of Engineering and Technology, BPUT, Bhubaneswar, Odisha	7.9/10	2018
B.Tech (Electronics and Instrumentation Engineering)	Institute of Technical Education and Research, Bhubaneswar, Odisha	7.6/10	2016
Odisha Board **(XII)**	Royal College of Science and Technology, Bhubaneswar, Odisha	56.33%	2012
Odisha Board **(X)**	Secondary Board High School, Cuttack, Odisha	73.33%	2010

Declaration

I hereby declare that all above-mentioned information given by me is true and correct to best of my knowledge and belief.

Sonali Mishra